

# The sharp-interface and phase-field cell problems are the same problem

## 1 Setup

A periodic cell  $\Omega$  splits into two subdomains meeting on an interface  $\Gamma$ ,

$$\Omega = \Omega_1 \cup \Omega_2, \quad \Omega_1 \cap \Omega_2 = \emptyset, \quad \Gamma = \partial\Omega_1 \cap \partial\Omega_2,$$

with constant conductivities  $K_1, K_2 > 0$  and  $\hat{\mathbf{n}}$  the unit normal on  $\Gamma$  pointing out of  $\Omega_1$ . Homogenization gives the effective tensor  $\mathbf{K}_{\text{eff}}$  from a *corrector* (cell) problem posed on  $\Omega$ . There are two ways to write that problem, and this note shows they are the same one. Nothing below uses any property of the phases beyond  $K_1, K_2$  being positive constants.

**Notation.**  $\chi$  is the *indicator function* of  $\Omega_1$ : the function equal to 1 inside  $\Omega_1$  and 0 outside,

$$\chi(\mathbf{x}) = \mathbf{1}_{\Omega_1}(\mathbf{x}) = \begin{cases} 1, & \mathbf{x} \in \Omega_1, \\ 0, & \mathbf{x} \in \Omega_2. \end{cases} \quad (1)$$

$\varphi$  is the phase field: a smooth function equal to 1 in  $\Omega_1$  and 0 in  $\Omega_2$ , sweeping between them across a band of decay length  $\varepsilon$ . It is a smoothed  $\chi$ .  $\langle \cdot \rangle = |\Omega|^{-1} \int_{\Omega}(\cdot)$ , and  $\llbracket \cdot \rrbracket$  is the jump across  $\Gamma$ .

**Problem A (sharp).** Find *two* cell-periodic fields,  $\mathbf{t}_1$  defined on  $\Omega_1$  only and  $\mathbf{t}_2$  defined on  $\Omega_2$  only, with

$$\nabla \cdot (K_1(\nabla \mathbf{t}_1 + \mathbf{I})) = \mathbf{0} \text{ in } \Omega_1, \quad \nabla \cdot (K_2(\nabla \mathbf{t}_2 + \mathbf{I})) = \mathbf{0} \text{ in } \Omega_2, \quad (2)$$

$$\mathbf{t}_1 = \mathbf{t}_2 \text{ on } \Gamma, \quad K_1(\nabla \mathbf{t}_1 + \mathbf{I}) \cdot \hat{\mathbf{n}} = K_2(\nabla \mathbf{t}_2 + \mathbf{I}) \cdot \hat{\mathbf{n}} \text{ on } \Gamma, \quad (3)$$

normalized by  $\int_{\Omega_1} \mathbf{t}_1 d\Omega + \int_{\Omega_2} \mathbf{t}_2 d\Omega = \mathbf{0}$ , and then

$$\mathbf{K}_{\text{eff}} = \frac{1}{|\Omega|} \left[ \int_{\Omega_1} K_1(\nabla \mathbf{t}_1 + \mathbf{I}) d\Omega + \int_{\Omega_2} K_2(\nabla \mathbf{t}_2 + \mathbf{I}) d\Omega \right]. \quad (4)$$

The two interface conditions are the physics: the field is continuous, and the normal flux is conserved. This is Calonne et al. (2011), Eqs. (7)–(11), whose  $\mathbf{t}_i$  (ice) and  $\mathbf{t}_a$  (air) are our  $\mathbf{t}_1$  and  $\mathbf{t}_2$ . In the jump notation, (3) reads  $\llbracket \mathbf{t} \rrbracket = \mathbf{0}$  and  $\llbracket K(\nabla \mathbf{t} + \mathbf{I}) \cdot \hat{\mathbf{n}} \rrbracket = \mathbf{0}$ , where the jump of a two-field quantity means (side 1)–(side 2). Note that A has two unknowns, B below has one; reconciling that is part of the claim (§2(f)).

**Problem B (phase field).** Find cell-periodic  $\mathbf{t}$  with

$$\nabla \cdot (K(\varphi)(\nabla \mathbf{t} + \mathbf{I})) = \mathbf{0} \text{ in } \Omega, \quad \langle \mathbf{t} \rangle = \mathbf{0}, \quad K(\varphi) = K_2 + (K_1 - K_2)\varphi, \quad (5)$$

$$\mathbf{K}_{\text{eff}} = \frac{1}{|\Omega|} \int_{\Omega} K(\varphi)(\nabla \mathbf{t} + \mathbf{I}) d\Omega. \quad (6)$$

One domain, one equation, no interface conditions.

**Claim.** A and B are the same problem. Two steps: §2 collapses A's two domains and two correctors into one weak equation for one field; §3 shows A's coefficient *is* B's coefficient. Neither step is a limit.

## 2 Step 1: two domains and two correctors become one weak equation

Let  $v \in H_{\text{per}}^1(\Omega)$  be an arbitrary cell-periodic scalar test function. We treat each subdomain separately, then add.

**(a) Test each bulk equation on its own subdomain.** Multiply the first equation of (2) by  $v$  and integrate over  $\Omega_1$ ; do the same on  $\Omega_2$ :

$$\int_{\Omega_1} v \nabla \cdot (K_1(\nabla \mathbf{t}_1 + \mathbf{I})) d\Omega = 0, \quad \int_{\Omega_2} v \nabla \cdot (K_2(\nabla \mathbf{t}_2 + \mathbf{I})) d\Omega = 0. \quad (7)$$

Both hold because the integrands vanish pointwise. No approximation yet.

**(b) Integrate by parts on each subdomain.** Using  $\int_D v \nabla \cdot \mathbf{A} = -\int_D \nabla v \cdot \mathbf{A} + \oint_{\partial D} v \mathbf{A} \cdot \hat{\mathbf{n}}_D$  with  $\hat{\mathbf{n}}_D$  the outward normal of  $D$ . The boundary of  $\Omega_1$  consists of  $\Gamma$ , where its outward normal is  $+\hat{\mathbf{n}}$ , and whatever part of the outer cell boundary  $\partial\Omega$  it touches; likewise for  $\Omega_2$ , whose outward normal on  $\Gamma$  is  $-\hat{\mathbf{n}}$ . Writing  $\mathbf{A}_m = K_m(\nabla \mathbf{t}_m + \mathbf{I})$  for  $m = 1, 2$ ,

$$0 = -\int_{\Omega_1} \nabla v \cdot \mathbf{A}_1 d\Omega + \int_{\Gamma} v \mathbf{A}_1 \cdot \hat{\mathbf{n}} d\Gamma + \underbrace{\int_{\partial\Omega \cap \partial\Omega_1} v \mathbf{A}_1 \cdot \hat{\mathbf{n}}_{\partial\Omega} dS}_{O_1}, \quad (8)$$

$$0 = -\int_{\Omega_2} \nabla v \cdot \mathbf{A}_2 d\Omega - \int_{\Gamma} v \mathbf{A}_2 \cdot \hat{\mathbf{n}} d\Gamma + \underbrace{\int_{\partial\Omega \cap \partial\Omega_2} v \mathbf{A}_2 \cdot \hat{\mathbf{n}}_{\partial\Omega} dS}_{O_2}. \quad (9)$$

The sign flip on the  $\Gamma$  term of (9) is the only subtlety:  $\Gamma$  is traversed with opposite orientation from the two sides.

**(c) Add the two equations.** This is the step that produces a single statement on  $\Omega$ . Adding (8) and (9),

$$0 = -\left[ \int_{\Omega_1} \nabla v \cdot \mathbf{A}_1 d\Omega + \int_{\Omega_2} \nabla v \cdot \mathbf{A}_2 d\Omega \right] + \int_{\Gamma} v (\mathbf{A}_1 - \mathbf{A}_2) \cdot \hat{\mathbf{n}} d\Gamma + O_1 + O_2. \quad (10)$$

Now dispose of the three groups in turn.

**(d) The outer boundary terms cancel.**  $O_1 + O_2$  is an integral over all of  $\partial\Omega$ . Both  $v$  and the fluxes  $\mathbf{A}_1, \mathbf{A}_2$  are cell-periodic, so they take equal values at corresponding points of opposite faces, where  $\hat{\mathbf{n}}_{\partial\Omega}$  points in opposite directions. Every contribution is therefore matched by its negative:  $O_1 + O_2 = 0$ .

**(e) The interface term is exactly the flux jump, which vanishes.** The middle term of (10) is  $\int_{\Gamma} v (\mathbf{A}_1 - \mathbf{A}_2) \cdot \hat{\mathbf{n}} d\Gamma$ , and  $(\mathbf{A}_1 - \mathbf{A}_2) \cdot \hat{\mathbf{n}} = 0$  on  $\Gamma$  is exactly the second interface condition in (3). So the term is 0 for every  $v$ , since the integrand vanishes pointwise.

(f) **The two volume integrals merge into one.** This is where  $K_\star$  and the single corrector  $\mathbf{t}$  come from. For both it is the same bookkeeping, with one extra check for  $\mathbf{t}$ . Define the piecewise-constant coefficient and the glued corrector

$$K_\star := K_1\chi + K_2(1 - \chi), \quad \mathbf{t} := \mathbf{t}_1\chi + \mathbf{t}_2(1 - \chi), \quad \text{so} \quad (K_\star, \mathbf{t}) = \begin{cases} (K_1, \mathbf{t}_1) & \text{on } \Omega_1, \\ (K_2, \mathbf{t}_2) & \text{on } \Omega_2. \end{cases} \quad (11)$$

Each piece of  $\mathbf{t}$  is only ever evaluated on its own subdomain, so  $\mathbf{t}_1$  never needs a value in  $\Omega_2$  or vice versa.

*The coefficient.*  $K_\star$  equals  $K_1$  at every point of  $\Omega_1$  and  $K_2$  at every point of  $\Omega_2$ . Nothing more is needed: a coefficient may jump, and  $K_\star$  does.

*The corrector.* The extra check concerns the gradient. On each open subdomain  $\nabla \mathbf{t} = \nabla \mathbf{t}_m$ , so the *piecewise* gradient of  $\mathbf{t}$  is  $\chi \nabla \mathbf{t}_1 + (1 - \chi) \nabla \mathbf{t}_2$ . The question is whether this is also the gradient of  $\mathbf{t}$  *as a function on*  $\Omega$ , i.e. its weak gradient, which is what a weak form on  $\Omega$  differentiates. Test against a smooth periodic vector field  $\boldsymbol{\psi}$  and integrate by parts on each subdomain exactly as in (b), with the same orientations and the same periodic cancellation as (d):

$$\int_{\Omega} \mathbf{t} \cdot \nabla \boldsymbol{\psi} \, d\Omega = - \int_{\Omega_1} \nabla \mathbf{t}_1 \cdot \boldsymbol{\psi} \, d\Omega - \int_{\Omega_2} \nabla \mathbf{t}_2 \cdot \boldsymbol{\psi} \, d\Omega + \int_{\Gamma} (\mathbf{t}_1 - \mathbf{t}_2) \boldsymbol{\psi} \cdot \hat{\mathbf{n}} \, d\Gamma. \quad (12)$$

The last term vanishes for every  $\boldsymbol{\psi}$  *if and only if*  $\mathbf{t}_1 = \mathbf{t}_2$  on  $\Gamma$ , the first interface condition in (3). When it does,  $\mathbf{t} \in H_{\text{per}}^1(\Omega)$  and its weak gradient is the piecewise one:

$$\nabla \mathbf{t} = \nabla \mathbf{t}_1 \text{ on } \Omega_1, \quad \nabla \mathbf{t} = \nabla \mathbf{t}_2 \text{ on } \Omega_2, \quad \text{with no part concentrated on } \Gamma. \quad (13)$$

Had the fields jumped,  $\nabla \mathbf{t}$  would carry an extra surface term  $[[\mathbf{t}]] \hat{\mathbf{n}} \delta_{\Gamma}$ . The piecewise integrals of (10) would not see it, but the weak form on  $\Omega$  would, so the two would no longer agree. This is the only place where  $K_\star$  and  $\mathbf{t}$  behave differently. The coefficient may be discontinuous; the corrector must not be, and continuity is exactly what permits the gluing.

*The merge.* With (11) and (13) we may substitute on each subdomain without changing either integral:

$$\int_{\Omega_1} \nabla v \cdot \mathbf{A}_1 + \int_{\Omega_2} \nabla v \cdot \mathbf{A}_2 = \int_{\Omega_1} \nabla v \cdot K_\star(\nabla \mathbf{t} + \mathbf{I}) + \int_{\Omega_2} \nabla v \cdot K_\star(\nabla \mathbf{t} + \mathbf{I}). \quad (14)$$

The two integrands are now the *same function* of  $\mathbf{x}$ , integrated over two disjoint sets whose union is  $\Omega$ . Additivity of the integral gives

$$\int_{\Omega_1} (\cdots) + \int_{\Omega_2} (\cdots) = \int_{\Omega_1 \cup \Omega_2} (\cdots) = \int_{\Omega} (\cdots), \quad (15)$$

the last equality because  $\Omega_1 \cup \Omega_2$  differs from  $\Omega$  only by  $\Gamma$ , a set of measure zero. So neither  $K_\star$  nor  $\mathbf{t}$  is assumed or fitted. Each is the name for “one thing on one piece and another on the other,” and (15) is the only thing done with them. The same additivity turns  $\mathbf{A}$ ’s normalization into  $\langle \mathbf{t} \rangle = \mathbf{0}$  and  $\mathbf{A}$ ’s  $\mathbf{K}_{\text{eff}}$  (4) into  $|\Omega|^{-1} \int_{\Omega} K_\star(\nabla \mathbf{t} + \mathbf{I})$ .

(g) **Result.** Substituting (d), (e), (f) into (10):

$$\boxed{\int_{\Omega} \nabla v \cdot K_\star(\nabla \mathbf{t} + \mathbf{I}) \, d\Omega = 0 \quad \forall v \in H_{\text{per}}^1(\Omega).} \quad (16)$$

**Remark 1** (Where the two interface conditions went). *They were not dropped; each was absorbed somewhere. Flux conservation  $\llbracket K(\nabla \mathbf{t} + \mathbf{I}) \cdot \hat{\mathbf{n}} \rrbracket = 0$  was used in step (e), and it is the natural condition of (16): run (e) backwards and any weak solution satisfies it automatically. Continuity  $\mathbf{t}_1 = \mathbf{t}_2$  was used in step (f), to glue the two correctors into one  $\mathbf{t} \in H_{\text{per}}^1(\Omega)$  via (12). Conversely, it is built into (16): a single  $H^1$  field on  $\Omega$  cannot jump, so restricting any weak solution to  $\Omega_1$  and  $\Omega_2$  returns a pair  $(\mathbf{t}_1, \mathbf{t}_2)$  with matching traces. Two correctors in A and one in B is therefore not a difference between the problems. A's pair is B's single field, written in two pieces.*

**Remark 2** (Why one arbitrary  $v$  on  $\Omega$  is not an extra assumption). *It may look as though testing with a single  $v$  defined on all of  $\Omega$  smuggles in something. It does not, and the step reverses: given (16), choose  $v$  supported strictly inside  $\Omega_1$ . Then the integral is over  $\Omega_1$  only, where  $K_\star = K_1$  and  $\mathbf{t} = \mathbf{t}_1$ ; undoing the integration by parts (no boundary term, since  $v$  vanishes near  $\partial\Omega_1$ ) returns the first equation of (2). Symmetrically for  $\Omega_2$ . Allowing  $v$  to straddle  $\Gamma$  then returns the interface term, hence (3). So (16) is equivalent to (2)–(3), not weaker.*

**Remark 3** (Same algebra for the physical field). *Nothing in (a)–(g) used the specific form  $\nabla \mathbf{t} + \mathbf{I}$ . Replacing it by  $\nabla T$  throughout collapses the physical problem, with its own two temperatures  $T_1$  on  $\Omega_1$  and  $T_2$  on  $\Omega_2$ ,  $\nabla \cdot (K_m \nabla T_m) = 0$ ,  $T_1 = T_2$  and  $K_1 \nabla T_1 \cdot \hat{\mathbf{n}} = K_2 \nabla T_2 \cdot \hat{\mathbf{n}}$  on  $\Gamma$ , into  $\int_\Omega \nabla v \cdot K_\star \nabla T = 0$  for the glued  $T$ , identically. See §7.*

### 3 Step 2: the coefficient is already the phase-field interpolation

It is tempting to finish by saying “now replace  $K_\star$  by a smooth  $K(\varphi)$  that converges to it.” That would be an approximation, and would need a convergence theorem. No replacement is needed.

**Lemma 1.** *Let  $K(\cdot)$  be any function with  $K(0) = K_2$  and  $K(1) = K_1$ . Then*

$$K(\chi) = K_\star \quad \text{pointwise on } \Omega, \quad (17)$$

*exactly — no limit, no smallness of  $\varepsilon$ .*

*Proof.*  $\chi$  takes only the values 0 and 1. At  $\mathbf{x} \in \Omega_1$ ,  $\chi = 1$ , so  $K(\chi) = K(1) = K_1 = K_\star$ . At  $\mathbf{x} \in \Omega_2$ ,  $\chi = 0$ , so  $K(\chi) = K(0) = K_2 = K_\star$ . Those are all the points of  $\Omega$ .  $\square$

For the affine  $K$  of (5) it is three lines:

$$K_\star \stackrel{(11)}{=} K_1 \chi + K_2 (1 - \chi) = K_2 + (K_1 - K_2) \chi = K(\chi). \quad (18)$$

**What this means.** Problems A and B do not have two constitutive laws that agree asymptotically. They have *one* law  $K(\cdot)$ , evaluated at two different arguments:

$$\text{A: } K(\chi), \quad \text{B: } K(\varphi).$$

So Step 2 approximates the *geometry*, never the equations, and every difference between A and B traces to the single object  $\varphi - \chi$ . A second consequence: since  $\chi$  never takes an intermediate value, the *shape* of  $K$  between 0 and 1 is invisible to Problem A. Every admissible interpolation — arithmetic, harmonic, any other — coincides exactly on  $\chi$ . The choice can therefore matter only through the diffuse band (§5).

**Remark 4** (Why the corrector needs no Step 2).  $K_\star$  and  $\mathbf{t}$  are glued the same way in (f), but only  $K_\star$  reappears here. There is no “ $\mathbf{t}(\varphi)$ ” interpolating  $\mathbf{t}_1$  and  $\mathbf{t}_2$  across the band, and none is needed.  $K$  is data:  $B$  has to be told what coefficient to use at every point, including inside the band, and the lemma shows its prescription  $K(\varphi)$  reduces to  $A$ ’s on  $\chi$ .  $\mathbf{t}$  is the unknown:  $B$  solves for one field on all of  $\Omega$ , and inside the band it simply takes whatever values the equation produces. At finite  $\varepsilon$  the two sharp pieces  $\mathbf{t}_1, \mathbf{t}_2$  have no separate meaning inside  $B_\varepsilon$ , and nothing requires them to. After Step 1  $A$  also has a single unknown in the same space  $H_{\text{per}}^1(\Omega)$ , so the two problems differ only in their coefficient, and that difference is carried entirely by  $\varphi - \chi$ .

**Proposition 1.** *If  $\varphi = \chi$ , Problems  $A$  and  $B$  are the same boundary-value problem and give the same  $\mathbf{K}_{\text{eff}}$ , identically.*

*Proof.* Step 1 gives  $A \Leftrightarrow (16)$ . Lemma 1 gives  $K_\star = K(\chi)$ , so (16) is the weak form of (5). The two  $\mathbf{K}_{\text{eff}}$  formulas agree by (14)–(15) applied to the integrand  $K(\nabla \mathbf{t} + \mathbf{I})$ .  $\square$

## 4 Finite $\varepsilon$ : what is left over

In practice  $\varphi \neq \chi$ , so Proposition 1 does not apply verbatim. By Lemma 1 the entire residual difference is carried by  $\varphi - \chi$ . For the affine  $K$  this is exact:

$$\delta K := K(\varphi) - K_\star = (K_1 - K_2)(\varphi - \chi), \quad \text{supp } \delta K = B_\varepsilon := \{0 < \varphi < 1\}. \quad (19)$$

Both problems are uniformly elliptic on the same bounds,  $\min(K_1, K_2) \leq K \leq \max(K_1, K_2)$ , so both are well posed and both admit the Dirichlet minimum principle: for a unit direction  $\mathbf{e}$ ,

$$A(K) := \mathbf{e} \cdot \mathbf{K}_{\text{eff}}[K] \mathbf{e} = \min_{v \in H_{\text{per}}^1(\Omega)} E_K[v], \quad E_K[v] := \langle (\nabla v + \mathbf{e}) \cdot K(\nabla v + \mathbf{e}) \rangle. \quad (20)$$

Write  $K_1 \rightarrow K(\varphi)$ ,  $K_2 \rightarrow K_\star$  with minimizers  $v_\varepsilon, v_\star$ . Here  $v_\star$  is the *glued* sharp corrector of (11); it is because gluing puts it in the same space  $H_{\text{per}}^1(\Omega)$  as  $v_\varepsilon$  that each can be tested in the other’s energy. For *any*  $v$ ,  $E_{K(\varphi)}[v] - E_{K_\star}[v] = \langle \delta K |\nabla v + \mathbf{e}|^2 \rangle$ . Testing each minimizer against the other’s energy gives two inequalities which combine into

$$\langle \delta K |\nabla v_\varepsilon + \mathbf{e}|^2 \rangle \leq A_\varepsilon - A_\star \leq \langle \delta K |\nabla v_\star + \mathbf{e}|^2 \rangle. \quad (21)$$

Both sides are integrals over  $B_\varepsilon$  only. The left one uses only the corrector already computed, so (21) is an *a posteriori* error bar, not an asymptotic claim. With the logistic profile  $\varphi = \sigma(s/\varepsilon)$ ,  $\sigma(u) = (1 + e^{-u})^{-1}$ , in the signed distance  $s$  to  $\Gamma$ , and  $\int_0^\infty (1 - \sigma) du = \ln 2$ ,

$$\|\varphi - \chi\|_{L^1(\Omega)} = 2 \ln 2 \, \varepsilon |\Gamma| + \mathcal{O}(\varepsilon^2) \implies |A_\varepsilon - A_\star| \leq 2 \ln 2 \, (K_1 - K_2) M^2 \frac{|\Gamma|}{|\Omega|} \varepsilon + \mathcal{O}(\varepsilon^2), \quad (22)$$

with  $M^2 = \max_{B_\varepsilon} |\nabla v + \mathbf{e}|^2$ . First order, computable prefactor, and no H-convergence theorem required. (That theorem is the right citation only for a non-affine  $K$ , where (19) becomes an inequality.)

## 5 The first-order error is a surface excess

To first order a thin graded layer is equivalent to a sharp interface carrying two *surface excesses* per unit area. Transport tangential to  $\Gamma$  crosses the layer’s sublayers in parallel and sees an excess conductance; transport normal to it crosses them in series and sees an excess resistance:

$$\Sigma_t = \int_{-\infty}^{\infty} [K(\varphi) - K_\star] ds, \quad \Sigma_n = \int_{-\infty}^{\infty} \left[ \frac{1}{K(\varphi)} - \frac{1}{K_\star} \right] ds. \quad (23)$$

These two numbers are the only  $\mathcal{O}(\varepsilon)$  data the band presents. Both are elementary for the logistic profile, whose defining property here is the antisymmetry  $\sigma(-u) = 1 - \sigma(u)$ .

$\Sigma_t = 0$ , **exactly**.  $K$  is affine, so the integrand is  $(K_1 - K_2)[\sigma(u) - H(u)]$  with  $H$  the unit step, and antisymmetry makes  $\sigma - H$  an odd function:

$$\Sigma_t = (K_1 - K_2) \varepsilon \int_{-\infty}^{\infty} [\sigma(u) - H(u)] du = 0. \quad (24)$$

Not first order — zero. The diffuse cell has *exactly* the sharp cell's phase fraction, so the parallel response carries no error at any  $\varepsilon$ . The leading correction is  $\mathcal{O}(\varepsilon^2 \kappa)$  from interface curvature.

$\Sigma_n \neq 0$ . Here  $1/K$  is convex in  $\varphi$ , so the band short-circuits a series resistance. With  $p = \sigma(u)$ ,  $du = dp/[p(1-p)]$  and partial fractions,

$$\Sigma_n = \varepsilon \int_0^1 \left[ \frac{1}{K_2 + (K_1 - K_2)p} - \frac{1}{K_*} \right] \frac{dp}{p(1-p)} = -\varepsilon \left( \frac{1}{K_2} - \frac{1}{K_1} \right) \ln \frac{K_1}{K_2} < 0. \quad (25)$$

Negative excess resistance: the diffuse cell conducts too well normal to  $\Gamma$ , first order in  $\varepsilon$ , with no free constants. This is the quantitative form of the familiar statement that an arithmetic mixture rule is right in parallel and wrong in series.

**Corollary 1** (An interpolation with no first-order bias). *The harmonic law  $K_h(\varphi) = (\varphi/K_1 + (1 - \varphi)/K_2)^{-1}$  has  $1/K_h$  affine in  $\varphi$ , so the same antisymmetry gives  $\Sigma_n = 0$  — but then  $\Sigma_t \neq 0$ . No scalar law nulls both. The tensorial law*

$$\mathbf{K}(\varphi) = K(\varphi)(\mathbf{I} - \hat{\mathbf{n}} \otimes \hat{\mathbf{n}}) + K_h(\varphi) \hat{\mathbf{n}} \otimes \hat{\mathbf{n}}, \quad \hat{\mathbf{n}} = \nabla \varphi / |\nabla \varphi|, \quad (26)$$

*uses each where its excess vanishes, giving  $\Sigma_t = \Sigma_n = 0$  and  $\mathcal{O}(\varepsilon^2)$ . Both branches satisfy  $K(0) = K_2$ ,  $K(1) = K_1$ , so Lemma 1 and Proposition 1 still apply.*

This is available because the conductivity cell problem is post-processing on a frozen  $\varphi$ : its interpolation is decoupled from whatever the evolution model uses.

## 6 Worked example and test: the laminate

For layers of period  $L$  with phase fraction  $\bar{\varphi}$  and  $n_\Gamma$  interfaces per period,  $\mathbf{K}_{\text{eff}}$  is diagonal with  $k_{\parallel} = \langle K \rangle$  and  $k_{\perp} = \langle K^{-1} \rangle^{-1}$ . Equations (24)–(25) give

$$k_{\parallel}(\varepsilon) = K_2 + (K_1 - K_2)\bar{\varphi} \quad \text{exact for every } \varepsilon, \quad (27)$$

$$\langle K^{-1} \rangle_{\varepsilon} = \frac{\bar{\varphi}}{K_1} + \frac{1 - \bar{\varphi}}{K_2} - \frac{n_\Gamma \varepsilon}{L} \left( \frac{1}{K_2} - \frac{1}{K_1} \right) \ln \frac{K_1}{K_2}, \quad k_{\perp} = \langle K^{-1} \rangle_{\varepsilon}^{-1}. \quad (28)$$

Equation (28) is not a truncation: its only error is tail overlap between neighbouring interfaces,  $\mathcal{O}(e^{-L/4\varepsilon})$ , which against direct quadrature is  $5 \times 10^{-5}$  relative at  $\varepsilon = L/50$  and machine precision by  $\varepsilon = L/128$ .

In our application  $K_1/K_2 = 2.29/0.02 = 114.5$ , giving  $(K_2^{-1} - K_1^{-1}) \ln(K_1/K_2) = 234.959$  and, at  $\bar{\varphi} = \frac{1}{2}$ ,  $n_\Gamma = 2$ :

| $\varepsilon$ | $\langle K^{-1} \rangle$ | $k_{\perp}$ | bias   |
|---------------|--------------------------|-------------|--------|
| $L/50$        | 15.8200                  | 0.063211    | +59.4% |
| $L/128$       | 21.5471                  | 0.046410    | +17.0% |
| $L/512$       | 24.3005                  | 0.041151    | +3.8%  |
| sharp         | 25.2183                  | 0.039654    | —      |

**How to test it.** Equation (28) is *linear in  $\varepsilon$  in resistivity* and a hyperbola in conductivity. Fit and plot  $1/k_{\perp}$  against  $\varepsilon$ ; then the test has two predicted numbers and no fitted ones,

$$\text{intercept} = \frac{\bar{\varphi}}{K_1} + \frac{1 - \bar{\varphi}}{K_2}, \quad \text{slope} = \frac{n_{\Gamma}}{L} \left( \frac{1}{K_2} - \frac{1}{K_1} \right) \ln \frac{K_1}{K_2}. \quad (29)$$

Two ways to get a meaningless answer: evaluating (29) at the nominal rather than the *measured*  $\bar{\varphi}$  (the discrete field's mean is not exactly  $\bar{\varphi}$ ), and refining  $\varepsilon$  at fixed mesh, which confounds this bias with mesh convergence. The committed driver is `studies/keff_sharp_limit/verification/`.

## 7 Carrying this to the transient equation with a phase-change source

The evolution model solves, instead of (5), an energy balance of the form

$$\underbrace{\rho(\varphi)c(\varphi)}_{\text{storage}} \partial_t T = \underbrace{\nabla \cdot (K(\varphi) \nabla T)}_{\text{conduction}} + \underbrace{\rho_1 L \partial_t \varphi}_{\text{latent heat}}. \quad (30)$$

Three things change, and they are not equally benign.

**1. Conduction: nothing new.** The operator is the same  $K(\varphi)$  appearing in (5). Step 1 collapses the sharp two-domain heat equation in the same way, Lemma 1 applies unchanged, and (24)–(25) hold verbatim. So the evolution model carries the *same* first-order conduction bias — too conductive normal to  $\Gamma$  — and Corollary 1 is the same fix.

**2. Storage: a nonzero excess, from a quadratic in  $\varphi$ .** Repeating (23) for the volumetric heat capacity  $C = \rho c$ , the excess stored enthalpy per unit interface area is  $\Sigma_C = \int [C(\varphi) - C_{\star}] ds$ . If  $C$  is interpolated *affinely* in  $\varphi$  this vanishes exactly, by the same antisymmetry as (24). But interpolating  $\rho$  and  $c$  *separately* and multiplying, as (30) is written, makes  $C$  *quadratic*:

$$\rho(\varphi)c(\varphi) = \rho_2 c_2 + (\rho_2 \Delta c + c_2 \Delta \rho) \varphi + \Delta \rho \Delta c \varphi^2, \quad \Delta \rho = \rho_1 - \rho_2, \quad \Delta c = c_1 - c_2, \quad (31)$$

and  $\varphi^2$  is *not* antisymmetric about the interface. Its excess is exactly computable by the substitution used for (25):

$$\int_{-\infty}^{\infty} [\sigma(u)^2 - H(u)] du = -1 \quad \implies \quad \Sigma_C = -\varepsilon \Delta \rho \Delta c \neq 0. \quad (32)$$

The diffuse cell stores *less* enthalpy than the sharp one, at first order in  $\varepsilon$  and proportional to the interfacial area. With  $\rho_1 c_1 = 1.80 \times 10^6$  and  $\rho_2 c_2 = 1.40 \times 10^3 \text{ J m}^{-3} \text{ K}^{-1}$ ,  $\Delta \rho \Delta c = 8.41 \times 10^5$ , so each interface loses the heat capacity of a  $0.47\varepsilon$ -thick slab of phase 1. On a cell with specific surface area  $|\Gamma|/|\Omega| = 2 \times 10^4 \text{ m}^{-1}$  that is a 1.9% deficit at  $\varepsilon = 1 \mu\text{m}$  and 9.3% at  $5 \mu\text{m}$ . *The fix is free*: interpolate the product,  $C(\varphi) = \rho_2 c_2 + (\rho_1 c_1 - \rho_2 c_2) \varphi$ , which is affine and has  $\Sigma_C = 0$  exactly. The two agree at  $\varphi \in \{0, 1\}$ , so Lemma 1 is indifferent between them — this is precisely a case where the interpolation is invisible to the sharp problem and matters only through the band.

**3. Latent heat: the integral is exact, the placement is not.** In the sharp model latent heat is a *surface* source on  $\Gamma$ , entering as a Stefan jump condition  $\llbracket K \nabla T \cdot \hat{\mathbf{n}} \rrbracket = \rho_1 L v_n$ . The phase-field model spreads it over the band as the volumetric  $\rho_1 L \partial_t \varphi$ . The total released is right:

$$\int_{\Omega} \rho_1 L \partial_t \varphi d\Omega = \rho_1 L \frac{d}{dt} \int_{\Omega} \varphi d\Omega = \rho_1 L \frac{dV_1}{dt}, \quad (33)$$

exactly, because  $\int_{\Omega} \varphi d\Omega$  is the exact phase-1 volume — the same  $\Sigma_t = 0$  antisymmetry as (24). What is *not* exact is where within the band it is deposited. Heat is injected across a layer of width  $\mathcal{O}(\varepsilon)$  over which  $T$  and  $K$  both vary strongly (here by a factor 114), so it enters at the wrong local temperature.

This third item is the one that needs care, and it is outside the scope of the analysis above: (21) and (23) assume a *frozen*  $\varphi$  and *no* source, which is exactly the situation of the  $\mathbf{K}_{\text{eff}}$  cell problem and exactly not the situation of (30). A moving diffuse interface carrying a source is the regime of the thin-interface analyses of Karma & Rappel and Almgren, whose known  $\mathcal{O}(\varepsilon)$  artifacts — modified interface kinetics, solute/thermal trapping, and a spurious surface diffusion along  $\Gamma$  — are generated by the source term and are most pronounced when the two phases' transport coefficients are very unequal, as here. Quantifying them requires a matched asymptotic expansion of (30), not the equilibrium excesses of §5.

**Remark 5** (The practical consequence). *A clean  $\varepsilon$  ladder for  $\mathbf{K}_{\text{eff}}$  certifies the conduction operator and the geometry representation. It does not certify the coupled model, which has two further  $\varepsilon$ -sensitivities: one cheap to remove ((32), by interpolating  $pc$  instead of  $\rho$  and  $c$ ), and one requiring its own analysis (the latent-heat source). Conversely, item 2 suggests its own test: the total enthalpy of a static two-phase cell should be  $\varepsilon$ -independent, which is the storage analogue of the  $k_{\parallel}$  gate and is equally cheap to run.*